Linear & Polynomial Regression

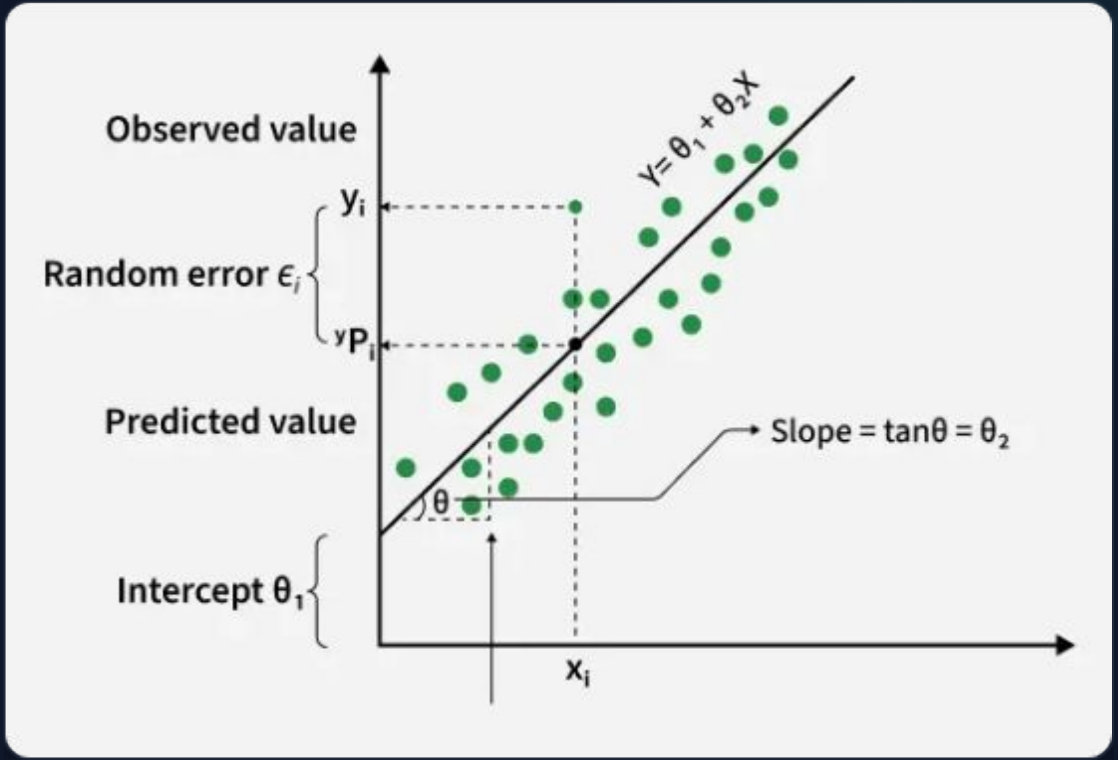
Comprehensive Guide to Predictive Modeling

Simple Linear Regression

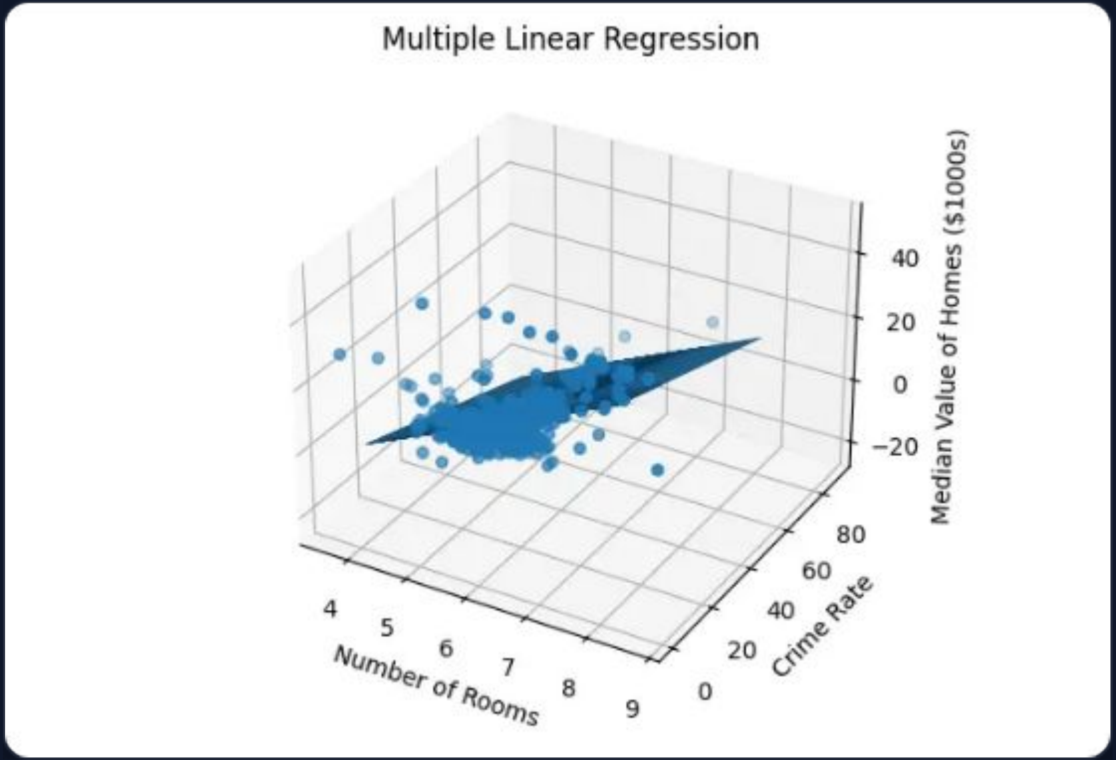
Simple Linear Regression (SLR) models the relationship between a single predictor (X) and a response (Y) using a straight line.

$$Y = \beta_0 + \beta_1 X + \varepsilon$$

- **β0 (Intercept):** Value of Y when X is zero.
- **β1 (Slope):** Change in Y per unit change in X.
- **ε (Error):** Random noise/unaccounted factors.



Multiple Linear Regression



Multiple Linear Regression (MLR) extends SLR by using two or more independent variables to explain the variance in Y.

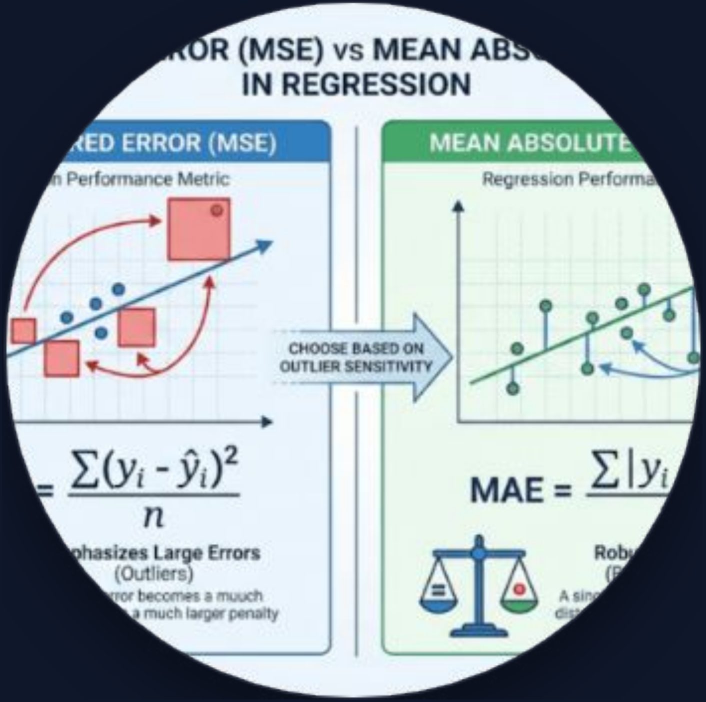
$$Y = \beta_0 + \beta_1 X_1 + \dots + \beta_n X_n + \varepsilon$$

Hyperplane Concept: Unlike a 2D line, MLR creates a multi-dimensional plane to fit the data points in higher-dimensional space.

Loss Functions

Quantifying Predictive Errors

Mean Squared Error (MSE)



Outlier Sensitivity

MSE squares the error, which quadratically penalizes large mistakes. This makes it highly sensitive to outliers, pulling the line of best fit towards them.

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

Mean Absolute Error (MAE)

Robustness

MAE takes the absolute value of differences, treating all errors linearly. It ignores the square of the distance, making it more robust to extreme values.

Optimization

While MSE optimizes the conditional mean, MAE focuses on the conditional median, making it ideal for skewed datasets.

Interpretability

MAE error is in the same units as the target variable, providing a direct "average mistake" interpretation.

MSE vs MAE Comparison

Feature	Mean Squared Error (MSE)	Mean Absolute Error (MAE)
Gradient	Varies with error size	Constant magnitude
Convergence	Stable near optimum	Can oscillate near zero
Outlier Impact	Heavy penalty (High)	Linear penalty (Low)
Statistical Focus	Conditional Mean	Conditional Median

Polynomial Regression

Capturing Non-Linear Relationships

Modeling the Curve

Polynomial regression transforms input features into higher degrees to fit non-linear data patterns while remaining a "linear" model in terms of parameters.

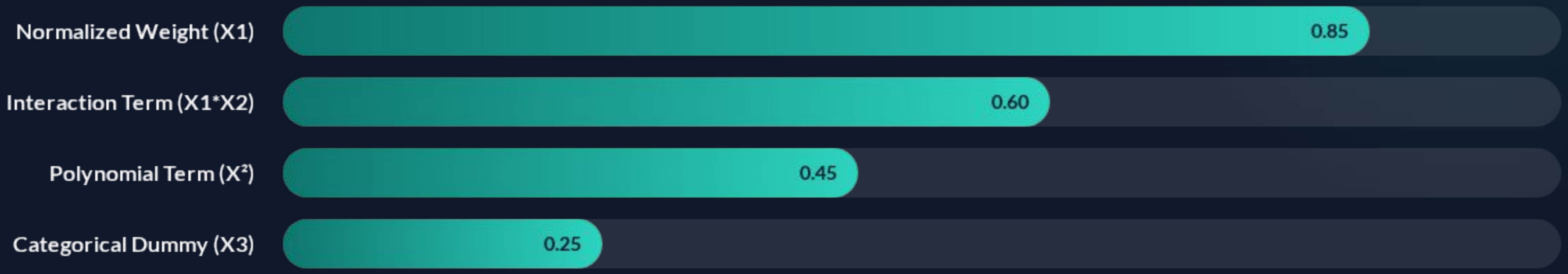
$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \ldots$$

Complexity Control: The degree of the polynomial determines the "bendiness" of the line, allowing it to capture peaks and troughs.

When to Apply Polynomial Features

- **Visible Curvature:** When exploratory scatter plots clearly show an arc, "U" shape, or periodic fluctuation that a straight line cannot capture.
- **Residual Patterns:** If a standard linear model shows systematic patterns (e.g., parabolic shapes) in its residuals, it indicates underfitting.
- **Domain Knowledge:** When theory suggests a physical or economic relationship is non-linear (e.g., quadratic growth in energy or area).
- **Warning:** Be cautious of high degrees ($d > 3$) as they risk severe **overfitting**, capturing random noise.

Feature Importance in Regression



Feature importance helps us identify which variables "drive" the prediction. We assess this using **standardized coefficients** or **p-values** to ensure we aren't just looking at the scale of the variable.

Q&A

Thank you for your engagement!

Visit documentation for code snippets & datasets.